

Secure Networking Architect

Certification

This exam is part of the following certification track:



Visit the [Cybersecurity Certification](#) page for information about certification requirements.

Exams available at [Pearson VUE](#) are listed below. Select an exam to view the details.

— Fortinet NSE 7 - Secure Networking 7.6.4 Architect

Status: ● Available

The Fortinet NSE 7 - *Secure Networking 7.6 Architect* exam evaluates your knowledge and expertise in designing, administering, and supporting secure SD-WAN and an enterprise security infrastructure composed of multiple FortiGate devices.

This exam tests your applied knowledge of advanced FortiGate configuration and operation, and includes operational scenarios; incident analysis; integration with FortiManager, FortiAnalyzer, and SD-WAN technologies; and troubleshooting scenarios.

Once you pass the exam, you will receive the following exam badge:



Audience

The Fortinet NSE 7 - *Secure Networking Architect* exam is intended for network and security professionals responsible for designing, administering, and supporting secure SD-WAN and an enterprise security infrastructure composed of multiple FortiGate devices.

Exam Details

Exam name	Fortinet NSE 7 - <i>Secure Networking Architect</i>
Time allowed	60–70 minutes
Exam questions	40–50 questions
Scoring	Pass or fail. A score report is available from your Pearson VUE account.

[Help](#)

Language	English
Product version	FortiGate 7.6, FortiManager 7.6, FortiAnalyzer 7.6

Exam Topics

Successful candidates have applied knowledge and skills in the following areas and tasks:

▼ System configuration and SD-WAN setup (20–30% of the exam)

Tasks	Details
Implement the Fortinet Security Fabric	• Fabric Connectors versus external connectors • Automation Stitches • Use cases—SAML single sign-on (SSO) in the Security Fabric, automated quarantine with Security Fabric and indicator of compromise (IoC) detection • Use cases—integrating FortiNAC with dynamic firewall addressing, adding FortiNDR (formerly FortiAI) to the Security Fabric • Use cases—FortiGate automation for configuration backups, running CLI scripts for high CPU scenarios
Configure different operation modes for a high availability (HA) cluster	• FortiGate Clustering Protocol (FGCP) • Active-active load balancing • Extension of FGCP—virtual clustering • Virtual MAC addresses • Ethernet types and sync optimization • Use cases—VDM partitioning (high traffic volume) • FortiGate Session Life Support Protocol (FGSP) • Sample FGSP standalone sync • Standalone sync—coverage and limits • Use cases—session synchronization encryption using IPsec tunnels, inspection with asymmetric traffic—layer 2, inspection with asymmetric traffic for cloud environments • Effective HA: FGCP, FGSP, VRRP insights
Implement enterprise networks using VLANs and VDOMs	• VLANs on FortiGate • Virtual LAN switch • VDOM types • Use cases—segmentation through VLANs, internet access through inter-VDOM routing
Deploy an enterprise SD-WAN setup	• SD-WAN fundamentals • SD-WAN basic components • Architecture components • Use case identification • SD-WAN direct internet access (DIA): DIA topologies, DIA best practices and recommended settings • Use cases—basic SD-WAN DIA setup • SD-WAN basic monitoring • SD-WAN traffic distribution and member health • SD-WAN widgets • SD-WAN traffic logs and events

▼ Central management (15–25% of the exam)

Tasks	Details
Implement branch configuration deployments	• Zero-touch provisioning (ZTP) of SD-WAN branches • ZTP basics • Device deployment with ZTP • Device blueprints and using CSV files to import devices

Tasks	Details
Use SD-WAN Manager and overlay orchestration	• FortiManager features for SD-WAN • SD-WAN management on FortiManager • Metadata variable configuration and use • SD-WAN core settings on FortiManager • Planning SD-WAN deployment with FortiManager • Key deployment elements • New SD-WAN deployments with a dedicated FortiManager • Converting a topology for SD-WAN with FortiManager • Templates and template groups • IPsec configuration with FortiManager • Describing IPsec templates available on FortiManager • Using IPsec templates to configure hub-and-spoke IPsec VPNs • Configuring IPsec interfaces as SD-WAN members • SD-WAN overlay template—FortiManager SD-WAN overlay template • Use cases

▼ Security profiles (5–15% of the exam)

Tasks	Details
Manage SSL/SSH inspection profiles	• Multiple clients connecting to multiple servers versus protecting SSL servers • SSL strategies—certificate inspection versus full inspection • Server certificate server name indication (SNI) check feature • Dealing with false positive events • HTTP and HTTPS code injection • Use cases—certificate errors on FortiGate
Use a combination of web filtering, application control, intrusion prevention system (IPS), and Internet Service Database (ISDB) to secure the network	• Impact on firewall performance • Security profiles—device performance • Use cases: protecting server and client targets, false positive events, HTTP and HTTPS code injection, IPS sensors using CVE patterns

▼ Rules and routing (25–35% of the exam)

Tasks	Details
Implement OSPF to route enterprise traffic	• OSPF overview, access lists, prefix lists, route maps, protocol redistribution • OSPF over IPsec, OSPF equal-cost multi-path (ECMP) • Use cases
Implement BGP to route enterprise traffic	• BGP overview • Access lists, prefix lists, route maps, protocol redistribution • ECMP with BGP routes • Loopback interfaces as BGP sources • The neighbor-group command • Optimizing BGP for rapid convergence: BGP convergence, route reflectors, BFD parameter, the graceful-restart command • Use cases

Tasks	Details
Design SD-WAN rules	• User-defined SD-WAN rules • SD-WAN rule lookup process • SD-WAN for local-out traffic • Implicit SD-WAN rule • Monitoring SD-WAN rule status • SD-WAN rule strategies • SD-WAN rule traffic matching criteria • Application steering and application learning phases • Internet services as destination criteria • Preferred member election based on strategy • Advantage given to higher priority members
Configure SD-WAN routing	• Key routing principles in SD-WAN • Policy routes • Route lookup process • Member static routes • Static routes for zones • Member probe routes • Session tables • Different protocol states • Common session flags • Session reevaluation and triggers • Routing changes in SNAT sessions • Routing designing for SD-WAN • Routing considerations • Determining which type of routing and routing protocols to use

▼ **Advanced IPsec (25–35% of the exam)**

Tasks	Details
Implement IPsec VPN IKE version 2	• Mastering IPsec topology design • Best practices—Dead Peer Detection (DPD) modes • Impact of outbound network address translation (NAT) for IPsec interfaces with no IP addresses • OpenSSL for IPsec VPNs • IPsec aggregate for redundancy and traffic load balancing • Overlapping routes in remote VPNs • Maximum transmission unit (MTU) issues in IPsec tunnels • Network performance optimization: MTU, TCP maximum segment size (MSS), and IPsec fragmentation • IPsec networks with FortiManager IPsec templates • IPsec templates • IPsec templates autorouting • Metadata variables in IPsec templates • IPsec template single and dual hub-and-spoke topology • The VPN manager and IPsec templates • Hardware offload with IPsec encryption and decryption, forward error correction (FEC), the session NPU-Flag field • Dual-hub topologies • SD-WAN with dual-hub topologies • Dual-hub options in the SD-WAN overlay template • Configuration specifics for large topologies • BGP routing and self-healing • BGP advanced options for SD-WAN • Routing options for dual-hub topologies • SD-WAN self-healing • Multiregion topologies and large deployments • Use cases for SD-WAN multiregion topologies • Routing specifics for multiregion topologies • MSSP deployments with SD-WAN • VRF-aware overlays • Use cases
Configure IPsec multihub, multiregion, and large deployments	• Dual-hub topologies and use cases • Dual-hub options in the SD-WAN overlay template • Configuration specifics for large topologies • BGP routing and self-healing • BGP advanced options for SD-WAN • Routing options for dual-hub topologies • SD-WAN self-healing • Multiregion topologies and large deployments • Use cases—SD-WAN multiregion topologies • Routing specifics for multiregion topologies • Common MSSP deployments with SD-WAN • VRF-aware overlays

Tasks	Details
Implement ADVPN to enable on-demand VPN tunnels between sites	• ADVPN operation and requirements • ADVPN in a hub-and-spoke network • ADVPN shortcut negotiation • ADVPN on FortiManager • Fortinet auto-discovery VPN (ADVPN) using FortiGate or FortiManager • Hub-and-spoke topologies using ADVPN with IBGP dual hub-and-spoke topologies using ADVPN with IBGP and EBGp • ADVPN using the FortiManager VPN manager • ADVPN using FortiManager IPsec templates • SD-WAN support for ADVPN • ADVPN 2.0 overlay placeholders • Designing a network with SD-WAN and ADVPN • Shortcut timeout, delay for shortcut failback, dependent shortcuts • Alternative overlay and routing designs for ADVPN • BGP on loopback design • ADVPN without BGP route reflection • Dynamic BGP • Use cases—ADVPN 1.0 challenges, ADVPN 2.0

Training Resources

The following resources are recommended for attaining the knowledge and skills that are covered on the exam. The recommended training is available as a foundation for exam preparation. In addition to training, you are strongly encouraged to have hands-on experience with the exam topics and objectives.

- *Enterprise Firewall 7.6 Administrator* course and hands-on labs
- *SD-WAN 7.6 Enterprise Administrator* course and hands-on labs
- *FortiOS 7.6 Administrator* course and hands-on labs
- *FortiManager 7.6 Administrator* course and hands-on labs
- *FortiAnalyzer 7.6 Administrator* course and hands-on labs
- *FortiAnalyzer 7.6 Analyst* course and hands-on labs
- *FortiOS 7.6.0—Administration Guide*
- *FortiOS 7.6.3—Administration Guide*
- *FortiOS 7.6.0—New Features*
- *FortiOS 7.6.3—New Features*
- *FortiOS 7.6.0—CLI Reference*
- *FortiOS 7.6.3—CLI Reference*
- *FortiManager 7.6.0—Administration Guide*
- *FortiManager 7.6.3—Administration Guide*
- *FortiManager 7.6.0—CLI Reference*
- *FortiAnalyzer 7.6—CLI Reference*
- *SD-WAN—Deployment Guide*

Experience

- 3 years of experience with networking
- 3 years of experience with network security
- 2 years of hands-on experience with FortiGate
- 2 years of hands-on experience with FortiManager
- 2 years of hands-on experience with FortiAnalyzer

Exam Sample Questions

A set of sample questions is available from the Fortinet Training Institute. These questions represent the exam content in question type and content scope. However, the questions do not necessarily represent all the exam content, nor are they intended to assess your readiness to take the certification exam.

See the [Fortinet Training Institute](#) for the course that includes the sample questions

Examination Policies and Procedures

The Fortinet Training Institute recommends that you review the exam policies and procedures before you register for the exam. Access important information on the [Fortinet Training Institute Policies](#) page, and find answers to common questions on the [FAQ](#) page.

Questions?

If you have more questions about the NSE Certification Program, contact us through the [Fortinet Training Institute Helpdesk](#) page.